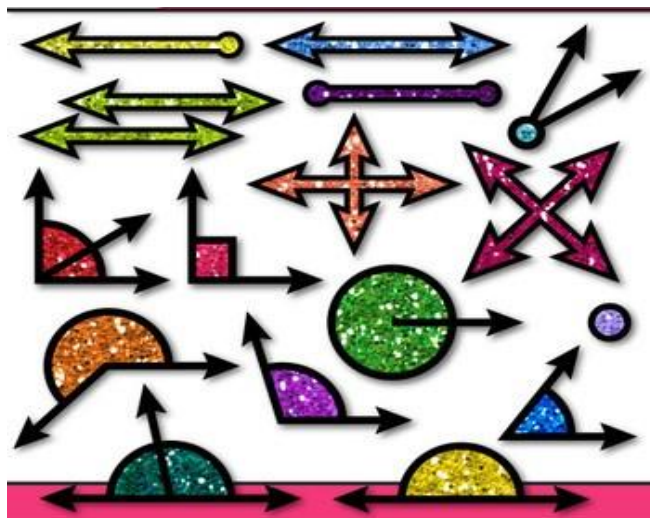


3

Lines and Angles



Introduction

Line Segment

It is the part of the line with two end points. The line segment AB is denoted by $\overline{AB}$.

Ray

It is a part of line which has a starting point and extends infinitely in one direction. The ray AB is denoted by $\overrightarrow{AB}$.

Angle

When two rays combined with a common end point, an angle is formed.

Collinear points

Three or more points which lie on the same line are called collinear points.

Non-Collinear points

Points that do not lie on one straight line are called non-collinear points.

Complementary angles

Two angles which sum upto 90° are called complementary angles.

Supplementary angles

Two angles which sum upto 180° are called supplementary angles.

Plane

A surface comprising all the straight lines that join any two points lying on it is called a plane.



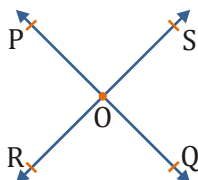
Do You Remember ?

- ★ Before starting the chapter you must recall the basic geometrical concepts that you studied in your previous classes.
- ★ You might have studied about line, line segment, ray, angles, types of angles etc.
- ★ In this chapter, we will extend out knowledge about pair of angles, specifically adjacent angles.

Intersecting lines & non-intersecting lines

Intersecting lines

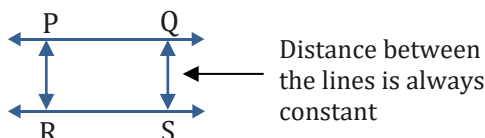
Two distinct lines are intersecting, if they have a common point. The common point is called the "point of intersection" of the two lines. We observe in figure, the intersection point of two intersecting lines is a unique point.



Non-Intersecting lines (Parallel lines)

Two distinct lines which are not intersecting are said to be parallel lines. The parallel lines are always at a constant distance from each other.

In figure, the lines PQ and RS do not intersect and hence, the lines PQ and RS are parallel to each other.

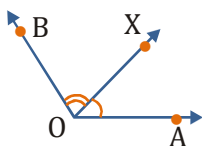


Pairs of angles

Adjacent angles

Two angles are called adjacent angles, if :

- They have the same vertex.
- They have a common arm, and
- Uncommon arms are on either side of the common arm.



In figure, $\angle AOX$ and $\angle BOX$ are adjacent angles. OX is common arm, OA and OB are non-common arms lies on the either side of OX .

When two angles are adjacent, then their sum is always equal to the angle formed by the two non-common arms. So, we can write, $\angle AOB = \angle AOX + \angle BOX$



In geometry, planes that do not intersect are called parallel planes."
For example- Books shelves

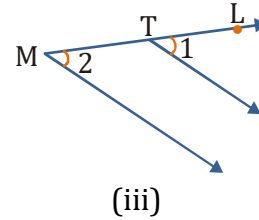
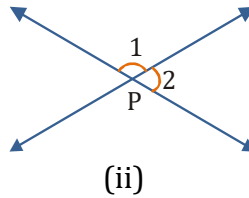
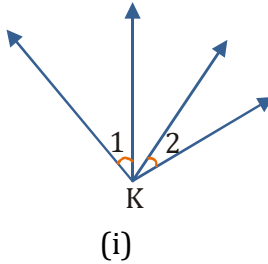
SPOT LIGHT



**Building
Concepts**

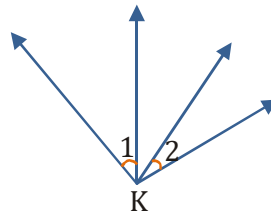
1

Determine whether $\angle 1$ and $\angle 2$ are adjacent angles.

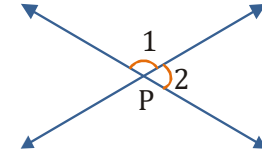


Explanation

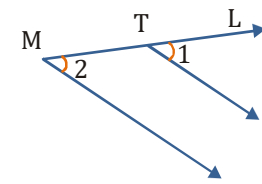
(i) No. They have the same vertex K, but no common arm.



(ii) Yes. They have the same vertex P and a common arm with no interior points in common



(iii) No. They do not have a common arm and a common vertex.



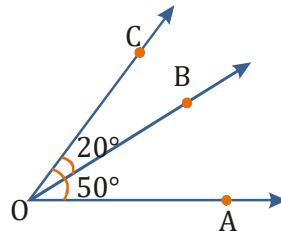
The side of $\angle 1$ is $\overrightarrow{TL}$; The side of $\angle 2$ is $\overrightarrow{ML}$



**Numerical
Ability**

1

In figure, $\angle AOC = 50^\circ$ and $\angle BOC = 20^\circ$, find $\angle AOB$.



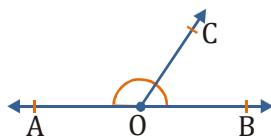
Solution

In figure, $\angle AOB$ and $\angle BOC$ are adjacent angles.

$$\begin{aligned} \Rightarrow \angle AOC &= \angle AOB + \angle BOC \\ \Rightarrow 50^\circ &= \angle AOB + 20^\circ \\ \Rightarrow \angle AOB &= 50^\circ - 20^\circ = 30^\circ \end{aligned}$$

Linear pair of angles

Two adjacent angles are said to form a linear pair of angles if non-common arms are two opposite rays.



In figure, the $\angle BOC$ and $\angle COA$ form a linear pair of angles because the non-common arms OA and OB are two opposite rays.

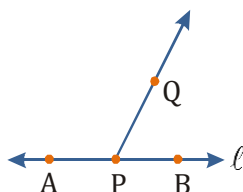
$$\angle AOC + \angle BOC = \angle AOB \quad [\angle AOC \text{ and } \angle BOC \text{ are adjacent angles}]$$

$$\therefore \angle AOC + \angle BOC = 180^\circ \quad [\text{Straight angle} = 180^\circ]$$

This result leads us to an axiom given below :

Axiom-1 : If a ray stands on a line, then the sum of two adjacent angles so formed is 180° .

This gives us another definition of linear pair of angles – when the sum of two adjacent angles is 180° , then they are called as linear pair of angles.



In figure, a ray PQ standing on a line ℓ forms a pair of adjacent angles $\angle BPQ$ and $\angle QPA$.

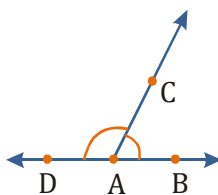
The sum of the measure of the two angles is a straight angle which is equal to 180° .

The ray PB and PA are opposite rays. Thus, the two angles form a linear pair of angles.

The above axiom can be stated in the reverse way as below:

Axiom-2 : If the sum of two adjacent angles is 180° , then the non-common arms of the angles form a line.

In figure, angles $\angle BAC$ and $\angle CAD$ are adjacent angles such that.



$$\angle BAC + \angle CAD = 180^\circ$$

Then, we have

$$\angle BAD = 180^\circ \quad (\angle BAD = \angle BAC + \angle CAD)$$

Thus, $\angle BAD$ is a straight angle.

Therefore, AB and AD are in one line.

The axiom 1 and 2 are called linear pair axioms. The two axioms are reverse of each other.



Be Alert !

- ★ We can use the linear pair axiom only when angles are being made on straight line i.e. straight line must be given.
- ★ In the questions where we need to prove that points are lying on a straight line, we are supposed to use converse of linear pair axiom.

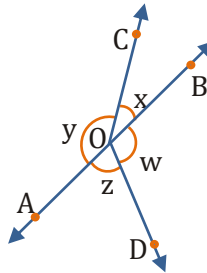


Numerical

Ability

2

In figure, if $x + y = w + z$, then prove that AOB is a line.



Solution

In this question, we can't use linear pair axiom since no straight line is given

$$\begin{aligned} \Rightarrow x + y &= w + z && \dots (1) \\ \Rightarrow x + y + w + z &= 360^\circ && [\text{Complete angle}] \\ \Rightarrow 2(x + y) &= 360^\circ \\ \Rightarrow x + y &= 180^\circ \end{aligned}$$

Now by converse of linear pair axiom, we can say that AOB is a straight line.

Also, since no straight line is given in the question, so we can't use vertically opposite angles.

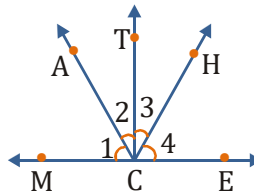


Building

Concepts

2

In the figure, $\overrightarrow{CM}$ and $\overrightarrow{CE}$ are opposite rays.



- Name the angle that forms a linear pair with $\angle 1$.
- Do $\angle 3$ and $\angle TCM$ form a linear pair? Justify your answer.

Explanation

- $\angle ACE$ and $\angle 1$ have common side $\overrightarrow{CA}$, the same vertex C, and opposite rays $\overrightarrow{CM}$ and $\overrightarrow{CE}$. So, $\angle ACE$ forms a linear pair with $\angle 1$.
- No, their non-common sides are not opposite rays.

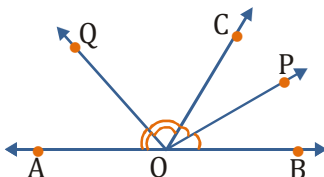


Building

Concepts

3

In figure, OP bisects $\angle BOC$ and OQ bisects $\angle AOC$. Prove that $\angle POQ = 90^\circ$.



Explanation

Given : In fig, OP bisects $\angle BOC$ & OQ bisects $\angle AOC$.

To Prove : $\angle POQ = 90^\circ$

Proof :

$$\angle POC = \frac{1}{2} \angle BOC \quad \dots (1) \quad [\text{OP bisects } \angle BOC]$$

$$\angle COQ = \frac{1}{2} \angle AOC \quad \dots (2) \quad [\text{OQ bisects } \angle AOC]$$

$$\angle POQ = \angle POC + \angle COQ$$

$$= \frac{1}{2} \angle BOC + \frac{1}{2} \angle AOC \quad [\text{From equation (1) \& (2)}]$$

$$= \frac{1}{2} (\angle BOC + \angle AOC)$$

$$= \frac{1}{2} \times 180^\circ \quad [\angle BOC + \angle AOC = 180^\circ] [\text{linear pair of angles}]$$

$$\Rightarrow \angle POQ = 90^\circ$$

Hence proved.

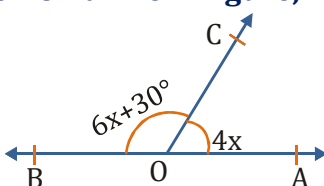


Numerical

Ability

3

What value of x would make AOB a line in figure, if $\angle AOC = 4x$ and $\angle BOC = (6x + 30^\circ)$?



Solution

If AOB is a line, then

$$\angle AOB = 180^\circ$$

[A straight angle = 180°]

$$\Rightarrow \angle AOC + \angle BOC = 180^\circ$$

$$\Rightarrow 4x + (6x + 30^\circ) = 180^\circ$$

$$\Rightarrow 10x + 30^\circ = 180^\circ$$

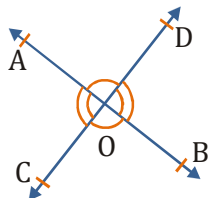
$$\Rightarrow 10x = 180^\circ - 30^\circ$$

$$\Rightarrow 10x = 150^\circ$$

$$\Rightarrow x = \frac{150^\circ}{10} = 15^\circ$$

Vertically opposite angles

If two lines intersect each other, then the pairs of opposite angles formed are called vertically opposite angles.



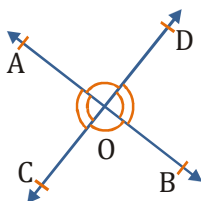
Two lines AB and CD intersect at point O.

Then, two pairs of vertically opposite angles are formed.

One pair is $\angle AOD$ & $\angle BOC$, the other pair is $\angle AOC$ & $\angle BOD$.

Theorem : If two lines intersect each other, then the vertically opposite angles are equal.

Given : Two lines AB and CD intersect at a point O.



Two pairs of vertically opposite angles are :

(i) $\angle AOC$ and $\angle BOD$

(ii) $\angle AOD$ and $\angle BOC$

To Prove : (i) $\angle AOC = \angle BOD$ (ii) $\angle AOD = \angle BOC$

i.e. two pairs of vertically opposite equal angles

Proof : CD is a straight line

$$\Rightarrow \angle AOC + \angle AOD = 180^\circ \quad \dots (1) \quad \text{[Linear pair]}$$

AB is a straight line

$$\Rightarrow \angle AOD + \angle BOD = 180^\circ \quad \dots (2) \quad \text{[Linear pair]}$$

From (1) & (2)

$$\Rightarrow \angle AOC + \angle AOD = \angle AOD + \angle BOD$$

$$\Rightarrow \angle AOC = \angle BOD$$

Similarly, we can prove that $\angle AOD = \angle BOC$.

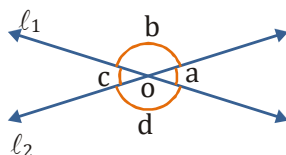


Numerical

Ability

4

In figure, lines ℓ_1 and ℓ_2 intersect at O, forming angles as shown in the figure. If $a = 35^\circ$, find the values of b, c and d.



Solution

Since, lines ℓ_1 and ℓ_2 intersect at O

$$\therefore \angle a = \angle c \quad [\text{Vertically opposite angles}]$$

$$\Rightarrow \angle c = 35^\circ \quad [\angle a = 35^\circ]$$

$$\text{Clearly, } \angle a + \angle b = 180^\circ \quad [\text{Linear pair of angles}]$$

$$\Rightarrow 35^\circ + \angle b = 180^\circ$$

$$\Rightarrow \angle b = 180^\circ - 35^\circ$$

$$\Rightarrow \angle b = 145^\circ$$

Since, $\angle b$ and $\angle d$ are vertically opposite angles.

$$\therefore \angle d = \angle b \Rightarrow \angle d = 145^\circ \quad [\angle b = 145^\circ]$$

$$\text{Hence, } \angle b = 145^\circ, \angle c = 35^\circ, \angle d = 145^\circ$$

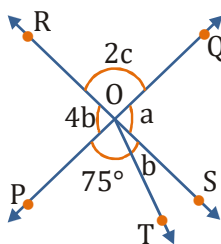


Numerical

Ability

5

In figure, two straight lines PQ and RS intersect each other at O. If $\angle POT = 75^\circ$, find the values of a, b and c.



Solution

Since, ROS is a straight line.

$$\therefore \angle ROP + \angle POT + \angle TOS = 180^\circ$$

$$\Rightarrow 4b + 75^\circ + b = 180^\circ$$

$$\Rightarrow 5b + 75^\circ = 180^\circ$$

$$\Rightarrow 5b = 180^\circ - 75^\circ$$

$$\Rightarrow 5b = 105^\circ$$

$$\Rightarrow b = \frac{105^\circ}{5}$$

$$\Rightarrow b = 21^\circ$$

Since, PQ and RS intersect at O.

$$\therefore \angle QOS = \angle POR \quad [\text{Vertically opposite angles}]$$

$$\Rightarrow a = 4b$$

$$\Rightarrow a = 4 \times 21^\circ = 84^\circ \quad [b = 21^\circ]$$

Since, ROS is a straight line.

$$\therefore \angle ROQ + \angle QOS = 180^\circ \quad [\text{Linear pair of angles}]$$

$$\Rightarrow 2c + a = 180^\circ$$

$$\Rightarrow 2c + 84^\circ = 180^\circ$$

$$\Rightarrow 2c = 180^\circ - 84^\circ$$

$$\Rightarrow 2c = 96^\circ$$

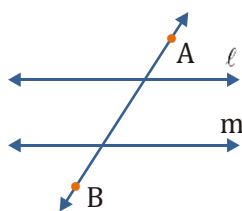
$$\Rightarrow c = \frac{96^\circ}{2}$$

$$\Rightarrow c = 48^\circ$$

Hence, $a = 84^\circ$, $b = 21^\circ$ and $c = 48^\circ$

Parallel lines and transversals

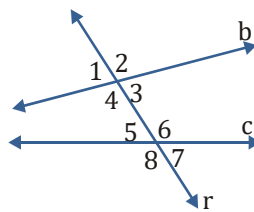
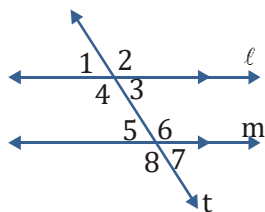
In geometry, a line, line segment or ray that intersects two or more lines at different points is called a transversal. $\overleftrightarrow{AB}$ is an example of a transversal. It intersects lines ℓ and m .



The lines cut by a transversal may or may not be parallel.

When a transversal intersects two lines, eight angles are formed, as shown in the figures.

These angles are given special names.



Parallel lines	Non-parallel lines
Interior angles lie between the two lines. $\angle 3, \angle 4, \angle 5, \angle 6$	Exterior angles lie outside the two lines. $\angle 1, \angle 2, \angle 7, \angle 8$
Alternate interior angles are on opposite sides of the transversal. $\angle 3$ and $\angle 5, \angle 4$ and $\angle 6$	Alternate exterior angles are on opposite sides of the transversal. $\angle 1$ and $\angle 7, \angle 2$ and $\angle 8$
Consecutive interior angles are on the same side of the transversal. $\angle 3$ and $\angle 6, \angle 4$ and $\angle 5$	Corresponding angles are on the same side of the transversal, having different vertices. One is an exterior angle and the other is an interior angle. $\angle 1$ and $\angle 5, \angle 4$ and $\angle 8$ $\angle 3$ and $\angle 7, \angle 2$ and $\angle 6$



Active
Maths

1

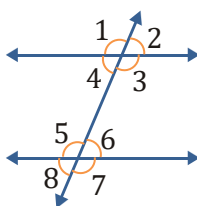
To investigate the relationship among the angles formed when a transversal intersects two parallel lines.

Materials : lined paper, scale, protractor

Exploring the concept

1. Use a scale to darken any two horizontal lines on a piece of lined paper.
2. Draw a transversal for the lines and label the angles 1 through 8. Use a protractor to find the measure of each angle.

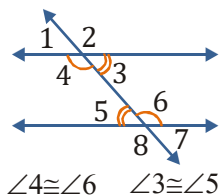
Drawing conclusions



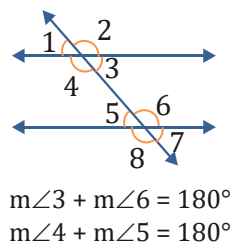
1. Compare the measures of the alternate interior angles & corresponding angles.
2. What is the sum of the measures of the consecutive interior angles?
3. Repeat Steps 1 and 2 above two more times by darkening different pairs of horizontal lines on your paper. Make the transversals intersect the lines at a different angle each time.
4. Do the interior angles relate to each other the same way for each pair of lines ?
5. Do the corresponding angles relate to each other the same way for each pair of lines ?
6. Compare the measures of the alternate exterior angles in each drawing.

The results of the activity suggest the theorems & axioms stated below.

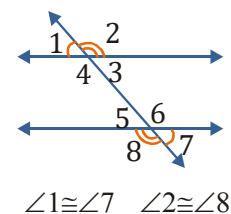
Theorem-1 : If two parallel lines are cut by a transversal, then each pair of alternate interior angles is congruent.



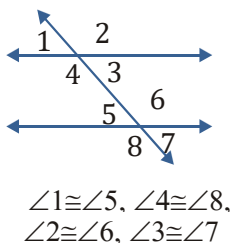
Theorem-2 : If two parallel lines are cut by a transversal, then each pair of consecutive interior angles is supplementary.



Theorem-3 : If two parallel lines are cut by a transversal, then each pair of alternate exterior angles is congruent.

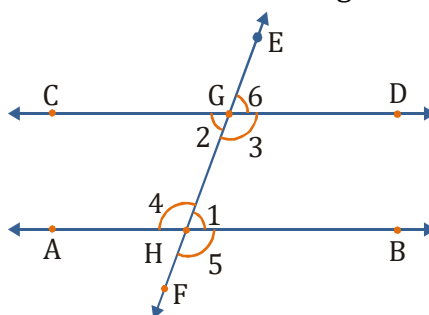


Axiom-1 : If two parallel lines are cut by a transversal, then each pair of corresponding angles is congruent.



Theorem : If a transversal intersects two parallel lines, then each pair of alternate interior angles is equal.

Given : AB and CD are two parallel lines and a transversal EF intersects them at point G and H respectively. Thus, the alternate interior angles are $\angle 2$ and $\angle 1$ and $\angle 3$ and $\angle 4$.



To Prove : $\angle 1 = \angle 2$ and $\angle 3 = \angle 4$

Proof :

$$\angle 2 = \angle 6$$

... (1) [Vertically opposite angles]

$$\angle 1 = \angle 6$$

... (2) [Corresponding angles]

From equation (1) and (2)

$$\angle 2 = \angle 6 \text{ and } \angle 1 = \angle 6$$

$$\Rightarrow \angle 1 = \angle 2$$

... (3)

Similarly, $\angle 4 = \angle 5$

... (4) [Vertically opposite angles]

$$\angle 3 = \angle 5$$

... (5) [Corresponding angles]

From equation (4) and (5)

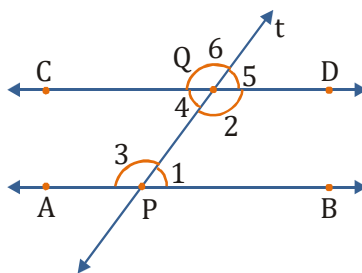
$$\angle 4 = \angle 5 \text{ and } \angle 3 = \angle 5$$

$$\Rightarrow \angle 3 = \angle 4$$

Hence, proved.

Theorem : If a transversal intersects two parallel lines, then each pair of consecutive interior angles are supplementary.

Given : AB and CD are two parallel lines. Transversal "t" intersects AB at P and CD at Q, making two pairs of consecutive interior angles, $\angle 1, \angle 2$ and $\angle 3, \angle 4$.



To Prove : (i) $\angle 1 + \angle 2 = 180^\circ$ and (ii) $\angle 3 + \angle 4 = 180^\circ$

Proof :

(i) $AB \parallel CD$

[Given]

$$\angle 1 = \angle 5 \quad \dots (1)$$

[Corresponding angles]

$$\angle 5 + \angle 2 = 180^\circ \dots (2)$$

[Linear pair]

From equation (1) and (2)

$$\angle 1 + \angle 2 = 180^\circ$$

(ii) $AB \parallel CD$

[Given]

$$\angle 3 = \angle 6 \quad \dots (4)$$

[Corresponding angles]

$$\angle 6 + \angle 4 = 180^\circ \dots (5)$$

[Linear pair]

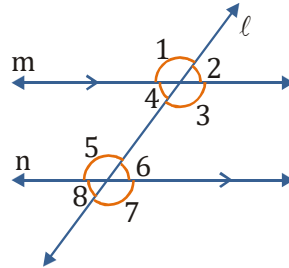
From equation (4) and (5)

$$\angle 3 + \angle 4 = 180^\circ$$

Hence, proved.



In figure, if $m \parallel n$ and $\angle 1$ and $\angle 2$ are in the ratio $3 : 2$, determine all the angles from 1 to 8.



Solution

Given $\angle 1 : \angle 2 = 3 : 2$

Let $\angle 1 = 3x^\circ$ and $\angle 2 = 2x^\circ$

$$\therefore \angle 1 + \angle 2 = 180^\circ \quad [\text{Linear pair of angles}]$$

$$\Rightarrow 3x^\circ + 2x^\circ = 180^\circ$$

$$\Rightarrow 5x^\circ = 180^\circ$$

$$\Rightarrow x^\circ = \frac{180^\circ}{5} = 36^\circ$$

$$\therefore \angle 1 = 3x^\circ = (3 \times 36)^\circ = 108^\circ$$

$$\text{And } \angle 2 = 2x^\circ = (2 \times 36)^\circ = 72^\circ$$

$$\angle 1 = \angle 3, \angle 2 = \angle 4 \quad [\text{Vertically Opposite Angles}]$$

$$\Rightarrow \angle 3 = 108^\circ, \quad \angle 4 = 72^\circ$$

$$\Rightarrow \angle 6 = \angle 2, \quad \angle 3 = \angle 7 \quad [\text{Corresponding angles}]$$

$$\Rightarrow \angle 6 = 72^\circ, \quad \angle 7 = 108^\circ$$

$$\angle 5 = \angle 7, \quad \angle 8 = \angle 6 \quad [\text{Vertically Opposite Angles}]$$

$$\Rightarrow \angle 5 = 108^\circ, \quad \angle 8 = 72^\circ$$

Hence,

$$\angle 1 = 108^\circ, \quad \angle 2 = 72^\circ, \angle 3 = 108^\circ, \angle 4 = 72^\circ,$$

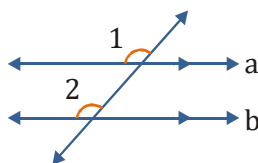
$$\angle 5 = 108^\circ, \quad \angle 6 = 72^\circ, \angle 7 = 108^\circ \text{ and } \angle 8 = 72^\circ.$$

Proving lines parallel

We can use geometry to prove that lines are parallel. We can also use geometry to construct parallel lines.

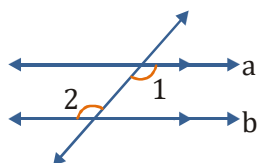
The following axioms & theorems illustrates five methods for proving that two lines are parallel.

Axiom : In a plane, if two lines are cut by a transversal so that a pair of corresponding angles is congruent, then the lines are parallel.



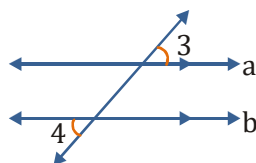
If $\angle 1 \cong \angle 2$, then $a \parallel b$

Theorem : In a plane, if two lines are cut by a transversal so that a pair of alternate interior angles is congruent, then the lines are parallel.



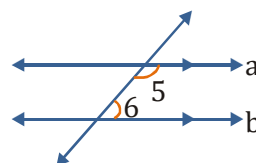
If $\angle 1 \cong \angle 2$, then

Theorem : In a plane, if two lines are cut by a transversal so that a pair of alternate exterior angles is congruent, then the two lines are parallel.



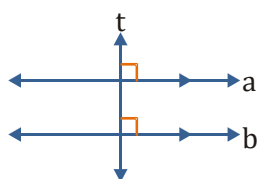
If $\angle 3 \cong \angle 4$, then $a \parallel b$

Theorem : In a plane, if two lines are cut by a transversal so that a pair of consecutive interior angles is supplementary, then the two lines are parallel.



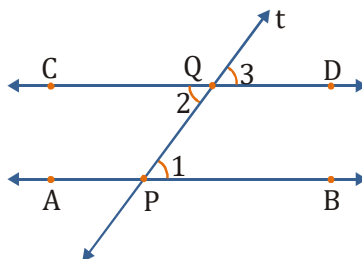
If $m\angle 5 + m\angle 6 = 180^\circ$ then $a \parallel b$

Theorem : In a plane, if two lines are perpendicular to the same line, then the two lines are parallel.



If $a \perp t$ and $b \perp t$, then $a \parallel b$

Theorem : If a transversal intersects two lines in such a way that a pair of alternate interior angles are equal, then the two lines are parallel.



Given : A transversal 't' intersects two lines AB and CD at P and Q respectively such that $\angle 1$ and $\angle 2$ are a pair of alternate interior angles and $\angle 1 = \angle 2$.

To Prove : $AB \parallel CD$

Proof :

$$\angle 2 = \angle 3 \quad \dots (1) \quad \text{[Vertically opposite angles]}$$

$$\angle 1 = \angle 2 \quad \dots (2) \quad \text{[Given]}$$

From equation (1) and (2)

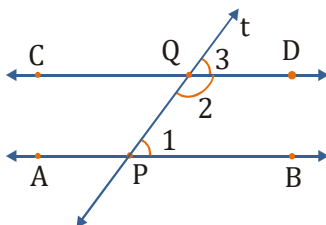
$$\angle 2 = \angle 3 \text{ and } \angle 1 = \angle 2$$

$$\Rightarrow \angle 1 = \angle 3$$

$AB \parallel CD$. [By converse of corresponding angles axiom]

Hence proved.

Theorem : If a transversal intersects two lines in such a way that a pair of consecutive interior angles are supplementary, then the two lines are parallel.



Given : A transversal t intersect two lines AB and CD at P and Q respectively such that $\angle 1$ and $\angle 2$ are a pair of consecutive interior angles, and $\angle 1 + \angle 2 = 180^\circ$.

To Prove : $AB \parallel CD$

Proof :

$$\angle 1 + \angle 2 = 180^\circ \quad \dots (1) \text{ [Given]}$$

$$\angle 2 + \angle 3 = 180^\circ \quad \dots (2) \text{ [Linear pair of angles]}$$

From equation (1) and (2)

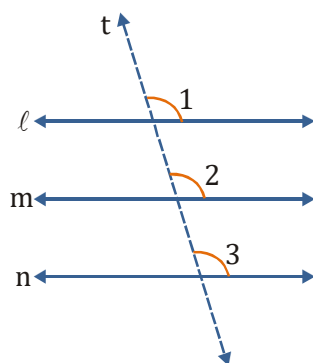
$$\angle 1 + \angle 2 = \angle 2 + \angle 3$$

$$\Rightarrow \angle 1 = \angle 3$$

$AB \parallel CD$ [By converse of corresponding angles axiom]

Hence proved.

Theorem : If two lines are parallel to the same line, they will be parallel to each other.



Given : $m \parallel l$ and $n \parallel l$.

To Prove : $m \parallel n$.

Construction : Draw a line t transversal for the lines l , m and n .

Proof :

$m \parallel l$ [Given]

$\therefore \angle 1 = \angle 2$... (1) [Corresponding angles]

$n \parallel l$ [Given]

$\therefore \angle 1 = \angle 3$... (2) [Corresponding angles]

From equation (1) and (2)

$\angle 1 = \angle 2$ and $\angle 1 = \angle 3$

$\Rightarrow \angle 2 = \angle 3$

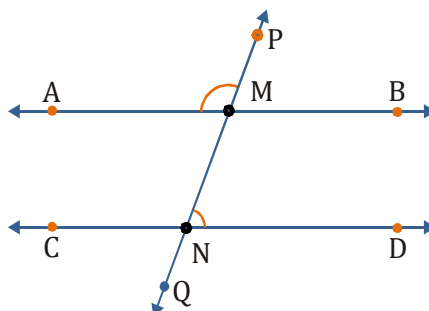
$\Rightarrow m \parallel n$ Hence proved [By converse of corresponding angles axiom]



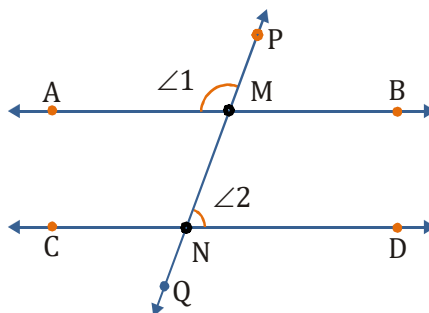
Quick Tips

- ★ It is advisable to draw neat and enlarged diagram while solving questions or doing proofs.
- ★ Angles can be named either using alphabets or by numbering them.
- ★ If angles are assigned numbers or numerical names then they must be shown clearly in the diagram.

For e.g.



The angles marked can be written as $\angle PMA$ and $\angle MND$. Or

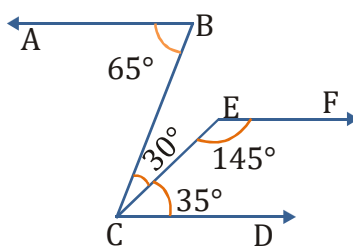


$\angle PMA$ can be shown as $\angle 1$.

$\angle MND$ can be shown as $\angle 2$.



In figure, $\angle ABC = 65^\circ$, $\angle BCE = 30^\circ$, $\angle DCE = 35^\circ$ and $\angle CEF = 145^\circ$. Prove that $AB \parallel EF$.



Explanation

$$\angle ABC = 65^\circ$$

$$\angle BCD = \angle BCE + \angle ECD = 30^\circ + 35^\circ = 65^\circ$$

$$\therefore \angle ABC = \angle BCD \quad [\text{Alternate Angles}]$$

$$\therefore AB \parallel CD \quad \dots (1)$$

$$\angle FEC + \angle ECD = 145^\circ + 35^\circ = 180^\circ$$

But these angles are consecutive interior angles formed on the same side of the transversal.

$$\therefore CD \parallel EF \quad \dots (2)$$

From (1) and (2),

$$AB \parallel EF$$

Hence proved.

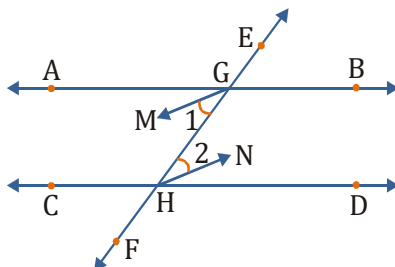


Building

Concepts

5

If two parallel lines are intersected by a transversal, then prove that the bisectors of any pair of alternate interior angles are parallel.



Explanation

Given : AB and CD are two parallel lines and transversal EF intersects them at G and H respectively. GM and HN are the bisectors of alternate angles $\angle AGH$ and $\angle GHD$, respectively.

To Prove : $GM \parallel HN$

Proof :

$$AB \parallel CD$$

[Given]

$$\therefore \angle AGH = \angle GHD$$

[Alternate interior angles]

$$\Rightarrow \frac{1}{2} \angle AGH = \frac{1}{2} \angle GHD$$

$$\Rightarrow \angle 1 = \angle 2$$

$$\Rightarrow GM \parallel HN \quad [\angle 1 \text{ \& \; } \angle 2 \text{ are alternate interior angles and are equal.}]$$

Hence, $GM \parallel HN$

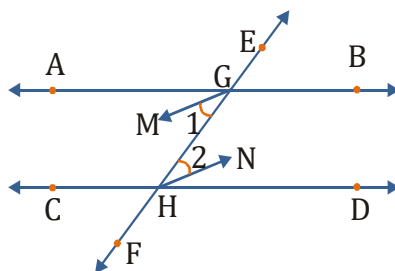


Building

Concepts

6

If the bisectors of a pair of alternate interior angle are parallel, then prove that given lines are parallel.



Explanation

Given : AB and CD are two straight lines cut by a transversal EF at G and H respectively.
GM and HN are the bisectors of alternate interior angles $\angle AGH$ and $\angle GHD$ respectively, such that $GM \parallel HN$.

To Prove : $AB \parallel CD$

Proof :

$$GM \parallel HN$$

[Given]

$$\therefore \angle 1 = \angle 2$$

[Alternate interior angles]

$$\Rightarrow 2\angle 1 = 2\angle 2$$

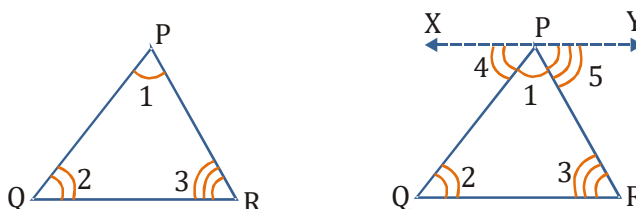
$$\Rightarrow \angle AGH = \angle GHD$$

$$\Rightarrow AB \parallel CD$$

[$\angle AGH$ & $\angle GHD$ are alternate interior angles formed by transversal EF with AB and CD, are equal.]

Angle sum property of a triangle

In previous classes, we have learnt that the sum of the three angles of a triangle is 180° . In this section, we shall prove this fact as a theorem.



Theorem : The sum of all the angles of a triangle is 180° .

Given : In a $\triangle PQR$,

$\angle 1$, $\angle 2$ and $\angle 3$ are the angles of $\triangle PQR$.

To Prove : $\angle 1 + \angle 2 + \angle 3 = 180^\circ$

Construction : Draw a line XPY parallel to QR through the opposite vertex P.

$$\text{Proof : } \angle 4 + \angle 1 + \angle 5 = 180^\circ$$

... (1)

[XPY is a line]

$$XPY \parallel QR$$

$$\therefore \angle 4 = \angle 2 \text{ and } \angle 5 = \angle 3$$

... (2)

[Alternate interior angles]

From equation (1) and (2)

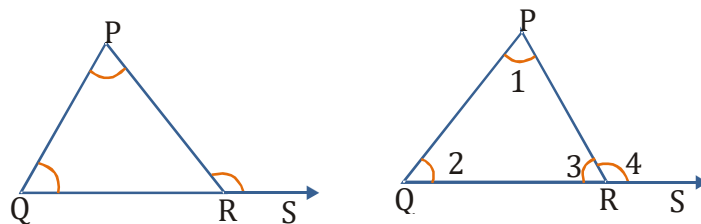
$$\angle 2 + \angle 1 + \angle 3 = 180^\circ$$

$$\text{or } \angle 1 + \angle 2 + \angle 3 = 180^\circ$$

Hence proved.

Exterior angle property of a triangle

Consider the $\triangle PQR$. If the side QR is produced to point S , then $\angle PRS$ is called an exterior angle of $\triangle PQR$. The $\angle PQR$ and $\angle QPR$ are called the two interior opposite angles of the exterior $\angle PRS$.



Theorem : If a side of a triangle is produced, then the exterior angle so formed is equal to the sum of the two interior opposite angles.

Given : In a $\triangle PQR$ $\angle 1$, $\angle 2$ and $\angle 3$ are the angles of $\triangle PQR$, side QR is produced to S and exterior angle $\angle PRS = \angle 4$.

To Prove : $\angle 4 = \angle 1 + \angle 2$

Proof :

$$\angle 3 + \angle 4 = 180^\circ \quad \dots (1)$$

[Linear pair of angles]

$$\angle 1 + \angle 2 + \angle 3 = 180^\circ \quad \dots (2)$$

[The sum of all the angles of a triangles is 180°]

From equation (1) and (2)

$$\angle 3 + \angle 4 = \angle 1 + \angle 2 + \angle 3$$

$$\Rightarrow \angle 4 = \angle 1 + \angle 2.$$

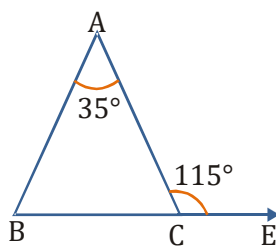
Hence proved.

**Numerical****7**

Ability

An exterior angle of a triangle is 115° and one of the interior opposite angles is 35° . Find the other two angles.

Solution



Let in $\triangle ABC$, exterior $\angle ACE = 115^\circ$ and $\angle A = 35^\circ$.

We know that,

$$\angle ACE = \angle A + \angle B$$

[Exterior angle is equal to sum of opposite interior angles]

$$\Rightarrow 115^\circ = 35^\circ + \angle B$$

$$\Rightarrow \angle B = 115^\circ - 35^\circ = 80^\circ$$

In $\triangle ABC$,

$$\angle A + \angle B + \angle C = 180^\circ \quad \text{[The sum of the three angles of a triangle is } 180^\circ]$$

$$\Rightarrow 35^\circ + 80^\circ + \angle C = 180^\circ$$

$$\Rightarrow 115^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 180^\circ - 115^\circ = 65^\circ$$

Hence, the other two angles are 80° and 65°

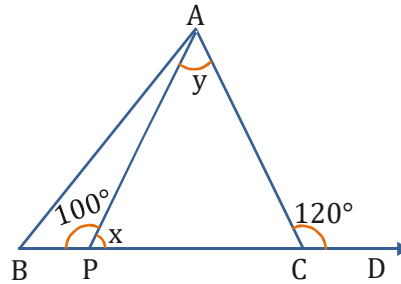


Numerical

Ability

8

In figure, $\angle ACD = 120^\circ$ and $\angle APB = 100^\circ$, find x and y .



Solution

$$x + 100^\circ = 180^\circ \quad [\text{Linear pair of angles}]$$

$$\Rightarrow x = 80^\circ$$

Now, we have

$$x + y = 120^\circ \quad [\text{Sum of interior opposite angles is equal to the exterior angle}]$$

$$\Rightarrow 80^\circ + y = 120^\circ$$

$$\Rightarrow y = 40^\circ$$

Hence, $x = 80^\circ$, $y = 40^\circ$

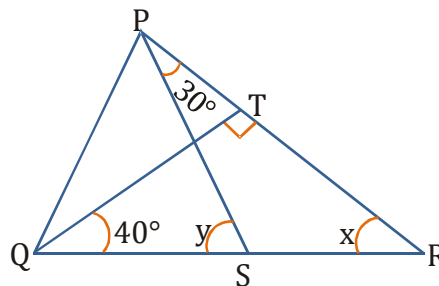


Numerical

Ability

9

In figure, if $QT \perp PR$, $\angle TQR = 40^\circ$ and $\angle SPR = 30^\circ$, find x and y .



Solution

In $\triangle TQR$, we have $\angle TQR + \angle QTR + \angle QRT = 180^\circ$

$$\Rightarrow 40^\circ + 90^\circ + x = 180^\circ$$

$$\Rightarrow 130^\circ + x = 180^\circ$$

$$\Rightarrow x = 50^\circ$$

Now, we have

$$y = x + 30^\circ \quad [\text{Sum of interior opposite angles is equal to the exterior angle}]$$

$$\Rightarrow y = 50^\circ + 30^\circ$$

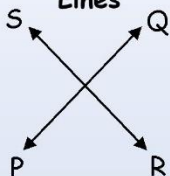
$$\Rightarrow y = 80^\circ$$

Hence, $x = 50^\circ$, $y = 80^\circ$

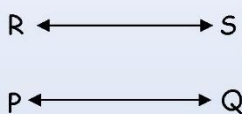
Memory map

Types of Lines

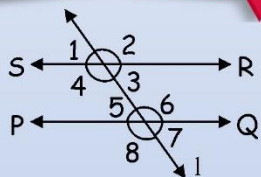
Intersecting Lines



Non-Intersecting Lines

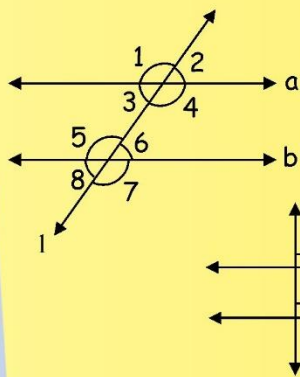


Parallel lines and transversal



- $\angle 4 = \angle 6$, $\angle 3 = \angle 5$ (Alternate interior angles)
 $\angle 3 + \angle 6 = \angle 4 + \angle 5 = 180^\circ$ (Co-interior angles)
 $\angle 1 = \angle 7$, $\angle 2 = \angle 8$ (Alternate exterior angles)
 $\angle 1 = \angle 5$, $\angle 4 = \angle 8 = \angle 2 = \angle 6$, $\angle 3 = \angle 7$ (corresponding angles)

Proving lines parallel



- If $\angle 1 = \angle 5$ then $a \parallel b$
 If $\angle 4 = \angle 5$ then $a \parallel b$
 If $\angle 2 = \angle 7$ then $a \parallel b$
 If $\angle 4 + \angle 6 = 180^\circ$ then $a \parallel b$

If $a \perp l$ and $b \perp l$ then $a \parallel b$

Types of Angles

Acute : Less than 90°

Right Angle : Equal to 90°

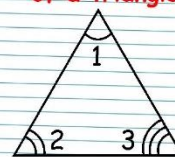
Obtuse : Angle between 90° and 180°

Pair of Angles

- Adjacent angles
- Linear pair of angles
- Vertically opposite angles
- Complimentary angles
- Supplementary angles

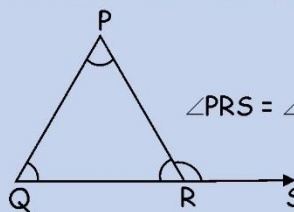
Triangle's Property

Angle sum property of a triangle



$$\angle 1 + \angle 2 + \angle 3 = 180^\circ$$

Exterior Angle of a Triangle



$$\angle PRS = \angle QPR + \angle PQR$$